Find the Percent of a Number

Lesson 4-4

Name: ________ Date: ________ Class: ________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

30% means 30 out of 100

Percent

A way to compare a number to 100, shown with the % sign.

In '25% of 80,' the base is 80

Base

The whole amount you take a percent of.

25% of 80 = 20; the part is 20

Part

The piece you get when you take a percent of a number.

part = percent × base

Equation

A math sentence with an equal sign showing both sides are the same.

$40 shirt at 25% off: discount = $10

Discount

Money taken off the first price to make it cheaper.

Key Ideas & Notes

- It's Discount Day at the arcade!
- Different games are offering percent-off deals on ticket prices.
- The claw machine is 25% off its usual 60-ticket price, and the racing game is 40% off 80 tickets.
- You need to calculate the discounts to figure out how many tickets each game actually costs!
- Find the percent of each number. Use the equation: Part = Percent × Base. Convert the percent to a decimal first.

Think About It

- What information do you need to find the discount amount?
- How is finding 25% of 60 different from finding 40% of 80?
- Is there more than one way to find a percent of a number?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

On Discount Day the claw machine is 25% off its 60-ticket price. In '25% of 60,' which number is the base (the whole), and what part are you trying to find?

Level 1 support

Start here: The base is 60 tickets and the part is the 25% discount we need to find.

The base is ____ and the part I want is ____.

La base es ____ y la parte que quiero es ____.

I need to find ____% of ____.

Necesito encontrar el ____% de ____.

Word bank: percent | base | part | discount

Listen for: *Students identify 60 as the base, the 25% discount as the part to find, and connect 'of' to the base.*

Level 2

Without computing exactly, will the discount be more or less than half of 60? Use a benchmark to estimate and explain.

- The discount will be ____ than half because ____.
- I estimate about ____ tickets because ____.

EXPLORE

What steps did you follow every time to find the percent of a number in the discount table?

Level 1 support

Start here: I changed the percent to a decimal, then multiplied it by the base to get the part.

I changed ___% into the decimal ___.

Cambié ___% al decimal ___.

Then I multiplied ___ x ___ = ___ (the part).

Luego multipliqué ___ x ___ = ___ (la parte).

Word bank: **percent** **base** **part** **equation**

Listen for: *Students describe the equation part = percent x base, converting the percent to a decimal first (e.g., 0.40 x 80 = 32).*

Level 2

Compare finding 25% of 60 with the decimal 0.25 versus dividing 60 into 4 equal parts. Why do both methods give the same part?

- Both give ___ because 25% is the same as ___.
- I prefer the ___ method because ___.

CONNECT

The arcade donates 12% of Saturday's 1,500 ticket sales. The manager says 180 tickets, a student says 18. Who is right, and how do you know?

Level 1 support

Start here: 12% of 1,500 is 0.12 times 1,500, which is 180 tickets.

The ____ is correct because 12% of 1,500 = ____ x 1,500 = ____.

El/La ____ tiene razón porque 12% de 1,500 = ____ x 1,500 = ____.

The mistake was using ____ instead of ____.

El error fue usar ____ en lugar de ____.

Word bank: **percent** **base** **part** **equation**

Listen for: *Students compute 0.12 x 1,500 = 180 and explain the student likely used 0.012 (divided by 1,000) instead of 0.12.*

Level 2

Estimate first: 10% of 1,500 is easy to find. Use that to check whether 180 is reasonable for 12%, and explain your reasoning.

- 10% of 1,500 is ____, so 12% should be a little ____.
- 180 is reasonable because ____.

Guided Examples

Example 1

What is 20% of 150?

Solution: 20% = 0.20. Multiply: $0.20 \times 150 = 30$.

Answer: A. 30

Example 2

What is 50% of 84?

Solution: 50% = 0.50. Multiply: $0.50 \times 84 = 42$. (Or simply divide by 2: $84 \div 2 = 42$.)

Answer: A. 42

Example 3

What is 10% of 320?

Solution: 10% = 0.10. Multiply: $0.10 \times 320 = 32$. (Or move the decimal one place left.)

Answer: A. 32

Write About the Math

The Writing Revolution

I can explain my steps using the words percent, base, part, and equation.

1. Kernel Sentence

subject + verb

Model: Percent is a way to compare a number to 100, shown with the % sign.
Porcentaje es una manera de comparar un número con 100, con el signo %.

Write a kernel sentence about percent. Use a subject and a verb.
Escribe una oración base sobre porcentaje. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Percent matters in math
Porcentaje importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Percent matters in math because ____.
Porcentaje importa en matemáticas porque ____.

___

but
pero

Percent matters in math, but ____.
Porcentaje importa en matemáticas, pero ____.

___

so
entonces

Percent matters in math, so ____.
Porcentaje importa en matemáticas, entonces ____.

___

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about percent.
Di un hecho verdadero sobre percent.

Percent ____.

Question
Pregunta

Ask a question about percent.
Haz una pregunta sobre percent.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about percent.
Muestra entusiasmo sobre percent.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with percent.
Dile a un compañero qué hacer con percent.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

The base is ____ and the part I want is ____.
La base es ____ y la parte que quiero es ____.

I need to find ____% of ____.
Necesito encontrar el ____% de ____.

I changed ____% into the decimal ____.
Cambié ____% al decimal ____.

Try It

Solve on your own. Check the answer key when you are done.

1. An arcade prize costs 240 tickets. If you have a coupon for 15% off, how many tickets do you save?

A. 36 tickets

B. 24 tickets

C. 15 tickets

D. 204 tickets

Show your work:

2. A student scored 72 out of 90 on a test. What percent did they score?

A. 80%

B. 72%

C. 75%

D. 90%

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A class has 32 students. 25% are in the math club, 50% are in the reading club, and 12.5% are in both. How many students are in each group? Can the total of all three groups be more than 32? Explain why or why not.

Sentence starter: There are ____ in math club, ____ in reading club, and ____ in both. The total ____ be more than 32 because ____.

Show your work:

Reflect — Exit Ticket

What is 45% of 360?

A. 162

B. 145

C. 180

D. 126

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 36 tickets *— 15% of 240 = 0.15 × 240 = 36 tickets saved.*
2. **Try It 2:** A. 80% *— 72 ÷ 90 = 0.80 = 80%.*
3. **Exit Ticket:** A. 162 *— 45% = 0.45. Multiply: 0.45 × 360 = 162.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Percent is a way to compare a number to 100, shown with the % sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).